

Mohit Baraiya

AI / ML Engineer

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SUMMARY

AI/ML Engineering student (B.E. AI & Data Science, GEC Rajkot) with hands-on experience building and deploying production LLM applications. Built SourceMind AI — a hybrid RAG chatbot combining dense + sparse retrieval on Pinecone, deployed on Render with FastAPI. Self-directed builder with a search → learn → build approach.

SKILLS

ML / Data: Data Preprocessing, EDA, Linear Regression, Logistic Regression, Decision Tree, Random Forest, K-Means, Ensemble Learning

Deep Learning: ANN, CNN, RNN, LSTM, GRU, Transformer Architecture — Keras, TensorFlow

Generative AI: RAG, Hybrid RAG, Pinecone, HuggingFace Embeddings, SemanticChunker, Groq API (LLaMA 3), LangGraph, Agentic AI (HITL, tool calling)

Languages: Python, SQL

Frameworks: FastAPI, LangChain

Database: Supabase (PostgreSQL)

Tools: Render, GitHub Pages, VS Code, Jupyter Notebook

Learning: LLM Evaluation (RAGAS), DSA in Python

EXPERIENCE

Flutter Developer Intern — Fuerte Developers

Jan 2025 – Jun 2025 | Rajkot, India

- Built Aapka Vyapar — a cross-platform business management app for small shopkeepers (sales, stock, purchases, customer records) using Flutter and Dart.
- Integrated Firebase authentication, Firestore real-time database, push notifications, and handled API integration and state management.

PROJECTS

SourceMind AI — Hybrid RAG Chatbot

[Live Demo](#)

Solo Project · Python · FastAPI · LangChain · Pinecone · HuggingFace · Groq (LLaMA 3) · Supabase

- Built a hybrid RAG pipeline combining dense HuggingFace embeddings with BM25 sparse vectors on Pinecone, using SemanticChunker parent-child chunking — significantly outperforms naive vector-only retrieval for PDF and YouTube transcript Q&A.
- Full user session isolation via UUID in localStorage; conversation memory and BM25 parameters persisted in Supabase across server restarts.
- Resolved Render free-tier memory crashes: singleton embedding model pattern, HuggingFaceEndpointEmbeddings, BM25 state moved to Supabase, cold-start wake-up ping.
- Backend on Render · Frontend on GitHub Pages · Supports PDF documents and YouTube transcripts.

Blueprint Agent — LangGraph Agentic Pipeline

[GitHub](#)

Solo Project · Python · FastAPI · LangGraph · Groq (LLaMA 3) · Tavily · Render · GitHub Pages

- Built an 8-node LangGraph state graph that turns a vague project idea into a downloadable PDF blueprint — covering market validation, system architecture, tech stack, and phase-wise build plan.
- Implemented Human-in-the-Loop (HITL) via LangGraph interrupt() — graph pauses mid-execution when clarification is needed and resumes after user answers; conditional routing skips HITL if idea is already clear.
- Validation node uses Tavily web search as a tool to check competitors, market fit, and unique angle. Stack recommendations are skill-aware (beginner / intermediate / advanced).
- Backend deployed on Render (FastAPI) · Frontend on GitHub Pages · PDF output with embedded SVG architecture diagram.

EDUCATION

Government Engineering College Rajkot

2025 – present

B.E. in AI and Data Science | Rajkot, India

R.K. University School of Diploma Studies

Completed 2025

Diploma in Computer Science & Engineering | Rajkot, India | SGPA: 8.90